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Bias and Accountability in AI Hiring Systems

INTRODUCTION

Artificial intelligence has grown significantly in importance and prevalence in our daily lives during the past few years. Artificial intelligence is now widely used by businesses in their recruiting and hiring procedures to screen applicants according to their own needs and preferences. Companies currently utilize AI techniques to analyze resumes, rank candidates, and filter out applications based on specific numerical scores within a framework called the "Application Tracking System" (ATS). Amazon's experimental AI hiring system, which was created around 2014 is a prime illustration of this. Their tool automatically evaluated resumes and identified top candidates. The system was trained on historical data collected over several years. Over time, it was discovered that the system showed a bias against female applicants. The algorithm penalized resumes containing words associated with women and favored resumes that reflected patterns common among previously hired male candidates. After discovering the ethical and technological issues, Amazon eventually discontinued utilizing it. In particular, this scenario shows how AI systems can reinforce pre existing workplace and social biases. Fairness, prejudice, accountability, openness, and equal opportunity are among the ethical issues this brings up.

The fact of the matter is that AI systems affect thousands of applications at very large scales, making the issue much more serious. Hiring decisions significantly influence career opportunities, income, and individuals' future success. Therefore, biased systems may unfairly prevent qualified applicants from receiving opportunities. To add to this context, this issue affects several groups starting with Job Applicants, Engineers and Developers, Companies, and society as a whole. Each stakeholder faces different risks, responsibilities, and consequences from using this system. This paper will examine the challenges associated with AI driven hiring systems through the lenses of Deontology, Virtue Ethics, and consequentialism. It discusses whether AI hiring systems should be used only with strict ethical oversight, complete transparency in the metrics used, and human supervision to maintain accountability.

BACKGROUND

To provide more context for this case, the details are explored as follows. Artificial intelligence and machine learning technologies have become commonplace in the workplace, specifically in the hiring and recruitment processes. These systems, while developed individually by companies, are made to reduce the manual work required of recruiters, human resources, and the concerned departments. As AI technology becomes more advanced and accessible, automated decision making is becoming more common across many industries. With respect to the specific case of Amazon, the timeline of events went as follows. Amazon started creating and testing a new AI hiring system around 2014. This project's objective was to automate some of the company's hiring procedures in order to lighten the workload for staff members. By evaluating resumes and allocating candidate rankings, the approach was intended to find the best

candidates. It was trained on historical data collected over 10 years. All resumes in this time period were analyzed, and the model was made based on this fact.

The bias emerged because the technology sector has historically been male dominated, especially in software engineering roles thus the system learned to recognize patterns shown by males. The quality and fairness of the system heavily relies on the quality and fairness of the training data so it began penalizing words and phrases often associated with women. A direct quote came from the Reuters article, in which a resume included the phrase "women's chess club captain," which was penalized. Although Amazon attempted to reduce the models bias after realizing the potential ethical concerns, they eventually discontinued its use. So, looking at it from an over the top perspective, the bias was not created intentionally, but it did arise from past data in which these biases existed.

Looking into the stakeholders, we start with applicants. Their interests and concerns are to expect the hiring system to provide fair and equal opportunities that determine their financial stability, growth, and success in life. Another major stakeholder in this situation is the engineering team. They are responsible for designing, training, and testing these hiring systems. They must ensure that systems operate accurately and ethically; they have the professional responsibility to reduce harmful bias. The last stakeholder is companies; they use AI systems to reduce recruitment time and hiring expenses, thereby making them more scalable and consistent. The last stakeholder would be society and government. Social concerns center on governments increasing focus on fairness and accountability in AI systems. Hiring discrimination can contribute to braoder social and economic inequality. Moreover, systems may unintentionally reinforce existing barriers that women and minorities face. To regulate and maintain ethical use

of AI standards, policymakers and regulators are beginning to explore laws and ethical standards for AI systems.

Finally, in a broader social context, AI decision making is spreading beyond hiring into areas such as healthcare, finance, law enforcement, education, and social media. As the particular field grows, the concerns surrounding it will grow. From all of this we can take that the larger ethical themes connected to the case include discussions of algorithmic accountability, workplace diversity, fairness in technology, transparency in artificial intelligence, and the role of human oversight in all of these aspects.

METHOD

Ethical frameworks play a key role in deciding whether, if engineering actions are ethically responsible or not. In this complex AI landscape, business needs to make a choice between technical innovation, accountability, efficiency, fairness, and equitable opportunity. Ethical approaches create different points of view and lead to a well informed and comprehensive choice of the most appropriate procedure. Within this paradigm we will specifically examine consequentialism, virtue ethics and deontological ethics.

Deontology

First, deontological ethics, in particular agent centered deontology, will be discussed. This is a framework that is based on responsibilities, obligations and values and not outcomes and actions. It implies that individuals should always be treated fairly and with respect in the eyes of all. On the other hand, it also considers some actions to be unethical, even if they lead to good results.

This case is a good fit for deontology, as Engineers and companies have a responsibility to ensure that the hiring system is fair and should be constructed in a way that does not discriminate against anyone based on gender. Companies have ethical obligations to the applicants because the hiring and selection process relates to people and their livelihoods. Even though it might improve efficiency, AI systems could ultimately be discriminatory in their approach to the people, and, therefore, be in breach of their moral responsibilities to people. In summary, this approach is useful for assessing companies' ethical responsibilities in hiring and recruitment with AI systems.

The Amazon case, in relation, unfairly penalized women which resulted in unequal treatment. Their actions were unethical, because they did not adhere to the principles of fairness and equal opportunity, which is a deontological perspective.

Virtue Ethics

Virtue ethics deals with moral virtues and the actions of people when they are making decisions. It highlights the personal attributes of honesty, integrity, fairness, accountability, responsibility and much more. It states that ethical behaviour is just as to do what a morally responsible person would do. It analyses the characters and motives of those making the decisions.

Engineers are key participants in the shaping of technologies in this particular case. An engineer's role is to show responsibility and integrity while designing an AI system, and this framework will help them understand if they did so in hiring systems or not. It also explores if companies show concern for moral duty or simply work to be effective and earn a profit.

However, a deeper analysis of the Amazon case revealed that the company ceased using the system eventually, accepting that the system had ethical issues.

Consequentialism

Consequentialism as a framework evaluates actions solely baked on the outcomes and consequences they produce. According to it, ethical decisions should maximize positive outcomes while minimizing harm. The morality of using it depends on the real world effects on individuals and society as a whole. To this effect, both short term and long consequences are important to evaluate decisions on the consequentialist perspective.

Consequentialism fits this case in the sense that AI hiring systems provide benefits of improving speed and efficiency, however at the same time they also can create significant harm for applicants and society as discussed in this paper. In Amazon's case they aimed to streamline their process, however they realised the discrimination that came with its use. They discontinued the system outweighing the discriminatory outcomes over the gains from it.

In general, it is understated how important it is to use multiple frameworks. First of all, no single ethical theory can adequately explain ethical problems. Different features are highlighted by each framework, providing a more comprehensive and balanced picture of the problem. In the following parts of this paper, these frameworks will be applied to the Amazon case to evaluate the responsibility of engineers and companies, judge the ethical concerns created by AI systems and weigh the broader social impact of using automated hiring systems.

ANALYSIS

The ethical concerns surrounding these systems can be examined differently depending on the ethical framework being applied. The Amazon system specifically demonstrates conflict between efficiency and fairness, automation and accountability and innovation and ethical responsibility. The above mentioned ethical frameworks will be used to evaluate the ethical concerns associated with AI hiring systems.

Ethical issue 1: Bias and discrimination

Looking in detail into the first ethical issue, we have bias and discrimination. The Amazon system developed bias against resumes associated with women, owing to learned patterns from historical training data reflected male prevalence in the industry. As a result of this, competent candidates were unjustly discriminated against on the basis of their gender rather than their aptitude for the position. It clearly illustrates how biases might be unintentionally created by such systems.

From a deontological perspective, companies and engineers have a duty to ensure hiring systems treat applicants fairly. For companies even if there is improved efficiency, if a biased system is in place it still is ethically incorrect. Now specifically looking at engineers, they have a responsibility to identify and reduce harmful bias during development and testing. From this framework's standpoint, in this case the biased hiring system was ethically problematic because it violates the principles of fairness and equal treatment.

Looking at the case from the perspective of virtue ethics, the framework emphasizes acting responsibly. Engineers should demonstrate, among other, honesty, fairness and accountability. In this case, engineers should actively investigate whether AI systems produce

harmful outcomes, and ignoring these demonstrates poor professional integrity. Companies should prioritize responsibility in addition to making profit to be the best possible version of themselves. With respect to Amazon, their decision to discontinue the system demonstrated recognition of the concerns involved.

From a consequential perspective, one should focus on real world consequences produced by AI hiring systems. Delving a little deeper into this, the positives from this system are faster hiring processes, reduced human workload, a greater ability to process larger numbers of applications, lowered hiring costs, increased consistency in hiring and potential reduction of human bias in recruitment. At the same time, the negatives from this system are discrimination especially against women and minority applicants, reinforcement of historical biases in the workplace, reduced trust in hiring system and AI technologies and potential legal and reputational issues for the company in question. Specifically with respect to Amazon, the aim of their system was to make their system streamlined and automated. However in the process of using this system, discriminatory outcomes were obtained which the company deemed to outweigh the potential gains, and as a result they discontinued the use of this system.

Ethical issue 2: Transparency and Accountability

Lack of transparency is a large issue at hand as many applicants do not understand the algorithm by which they are evaluated. The problem is exacerbated by the fact that they frequently get no reason for their rejection.

Deontologically speaking, candidates should be treated fairly and transparently throughout the employment process. When it comes to career decisions, companies have an ethical commitment to give accountability wherein this obligation is violated by a lack of

transparency. The perspective through virtue ethics is also very closely related to the deontological perspective. Where engineers should value openness, transparency and accountability, which in turn reflects integrity and ehtical leadership.

From a consequentialist perspective, the lack of transparency reduces the public trust in these systems, applicants lose confidence in the hiring process, and this reduced trust may negatively affect both companies and the society at large.

Ethical issue 3: Human oversight and accountability

AI systems make important decisions affecting people's lives, and the question arises about who should be responsible for these decisions. It can be engineers, companies as a whole, managers and/or developers.

From a deontological perspective, humans remain morally responsible using AI, so companies and engineers cannot avoid responsibility by blaming algorithms. It argues that ethical duties still apply to people at the company and the company as a whole even when decisions are automated. From the perspective of virtue ethics, engineers should maintain human oversight of systems affecting people. Responsible leadership means monitoring systems for harmful outcomes and accountability reflects strong integrity and professionalism. From a consequential perspective, poor oversight increases the likelihood of discrimination. Such systems in this case can create widespread social harm, thus to mitigate, a potential and viable solution could be to have human supervision to maintain trust.

Looking at the bigger picture, AI hiring systems improve efficiency, have faster recruitment, save on hiring costs, and are scalable for large companies. On the other hand, they can cause bias and discrimination, are not transparent, reduce accountability, reinforce inequality,

and reduce public trust and faith in such systems. Looking at this, the Amazon case demonstrates that technological efficiency alone does not guarantee good and ethical decision making and AI systems can unintentionally reinforce harmful societal biases if not carefully monitored. It implies that engineers and companies have ethical responsibilities when it comes to affecting people's opportunities and livelihoods, and oversight and transparency are key to maintaining trust in such systems in turn providing a fair and equal opportunity for all.

RECOMMENDATION

Given all that we have seen in this paper, here is my recommendation regarding the ethical issue at hand. In my opinion, AI hiring systems should not be banned, but should not operate without oversight or accountability. Companies should use systems with human supervision, transparency, regular bias testing and ethical oversight. To support this conclusion, AI systems improve efficiency and speed in recruitment and large companies will find it tedious to process thousands upon thousands of applicants. They assist recruiters by reducing repetitive work and reducing human inconsistencies. At the same time, these systems can unintentionally reproduce historical discrimination. Also added to this is the fact that they affect people's lives and future opportunities. So without monitoring, they may unfairly disadvantage qualified applicants and affect more than they were intended to do.

To inculcate all of these in a feasible system, I recommend that AI systems should assist human recruiters rather than fully replace them. Final decisions should involve human review which gives more accountability and reduces blind reliance on algorithms. To aid in this cause, companies should perform regular bias audits, wherein they routinely test AI systems for

discriminatory outcomes using diverse applicant data, and mitigate any negative consequences that come with it. Following this, they should also clearly communicate when and how AI systems are being used in these decisions, and in general focus toward achieving more accountability of the system. Finally Engineers should prioritize ethical responsibility during development, and they should report or address harmful bias when identified.

To achieve this there are limitations that stand in the way. To begin with, by design bias can never be fully eliminated from an AI system. Additionally, human processes also contain bias which add to this. Finally tracing off of this, continuous auditing increases costs and complexities to a company.

To conclude, AI hiring systems can provide important benefits when used responsibly, but this should not come at the cost of fairness and equal opportunity to all those that are applying. The Amazon case, while part of the larger context, demonstrates the importance of needing ethical oversight in engineering systems. Ultimately, AI should be used to support human decision making and analysis, rather than completely taking it over.

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